INCH POUND

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DETAIL SPECIFICATION
# INITIATOR, CARTRIDGE ACTUATED, M27 ASSEMBLY

*This specification is approved for use by all Departments and Agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This specification covers the minimum requirements for the M27 Cartridge Actuated Initiator Assembly (see 6.1).

## 2. APPLICABLE DOCUMENTS

**2.1 General.** The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

**2.2 Government documents.**

**2.2.1 Specifications, standards, and handbooks.** The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | |
|---|---|
| MIL-A-2550 | Ammunition, General Specification for |
| MIL-DTL-60517 | Initiator, Cartridge Actuated, M 27 Metal Parts Assembly |
| MIL-DTL-82763 | Initiator, Cartridge Actuated, M27, Remanufactured Metal Parts Assembly |

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to pennam@ih.navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/.

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AMSC N/A FSC 1377

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(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

**2.2.2 Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD (GAGE CODE 19200 and 19203)

| | |
|---|---|
| 8597230 | Cartridge, Impulse, M91 Assembly |
| DL10520550 | Initiator, Cartridge Actuated, M27, Assembly |

(Application for copies should be addressed to Commanding Officer, Naval Surface Warfare Center, Indian Head, Technical Information Branch (Code 4230), Indian Head, MD 20640.)

**2.3 Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E1742/E1742M Standard Practice for Radiographic Examination

(Copies of this document are available online at http://www.astm.org/cgi-bin/SoftCart.exe/STORE/standardsearch.shtml?L+mystore+afpt6301+1062008944 or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN SOCIETY FOR QUALITY

| | |
|---|---|
| ASQ Q9000 | Quality Management Systems – Fundamentals and Volcabulary |
| ASQ Z1.4 | Sampling Procedures and Tables for Inspection by Attributes |

(Application for copies should be addressed to the American Society for Quality Control, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

**2.4 Order of precedence.** In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 Materials.** All materials used in the manufacture of the initiators shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure there from has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder’s selections shall be clearly stated in the proposal.



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**3.2 Primary components.** For the purpose of this specification, the M91 cartridge (Drawing 8597230) is defined as a primary component. Only primary components from a single lot shall be used in the production lot of initiators. One primary component production lot may be used in more than one initiator production lot.

**3.3 Breakaway torque.** The chamber and cap of each assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation (see 4.3.1).

**3.4 Residual magnetism.** The initiator shall not deflect the indicator of the compass more than 5 degrees in either direction using procedures in 4.3.2.

**3.5 Firing pin release.** The firing pin of the assembled initiator, less safety pin and cartridge, shall release from the cocked position when the application of an axial load of 15 to 30 pounds has withdrawn the initiator pin 0.370 inch, minimum, and 0.425 inch, maximum (see 4.3.3).

**3.6 Initiator pin security.** Application of an axial load of 200 to 220 pounds to the initiator pin of the assembled initiator shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin (see 4.3.4).

**3.7 Functional.** (See 4.3.5)

**3.7.1 Ballistics.** Initiators loaded with cartridges from one accepted lot and temperature conditioned shall perform within the pressure values and ignition delay time (see 6.6.2) specified in table I. Pressure values shall be recorded at the end of 15 feet of aircraft hose, MS28741-4, and a pressure chamber with a volume of 1 cubic inch.

TABLE I. *Pressure values.*

| Temperature (°F) | Pressure (psi) | Ignition delay time Max (ms) |
|---|---|---|
| -60 to -75 | 700 to 1425 | 65 |
| 65 to 75 | 800 to 1550 | 65 |
| 195 to 205 | 900 to 1700 | 65 |

**3.7.2 Misfires.** There shall be no misfires.

**3.7.3 Mechanical failures.** The initiator shall function without mechanical failure (see 6.6.1).

**3.8 Workmanship.** Initiators shall be free of cracks, splits, incomplete or damaged threads on inlet and outlet ends, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

**3.8.1 Metal defects.** All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defects.

**3.8.2 Burr.** No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

**3.8.3 Foreign matter.** No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

**3.9 Radiographic examination.** Each initiator assembly shall be radiographically examined for defects as specified in 4.4.



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**3.10 Component parts.** All component parts shall meet the requirements of MIL-DTL-60517 or MIL-DTL-82763.

## 4. VERIFICATION

**4.1 Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. Conformation inspection.

**4.1.1 Quality assurance terms and definitions.** Reference shall be made to ASQ Q9000 to define the quality assurance terms used.

**4.1.2 Inspection.** Inspection shall be in accordance with MIL-A-2550 and ASTM E1742/E1742M.

**4.2 Inspection provisions.**

**4.2.1 Lot.**

**4.2.1.1 Submission of product.** Unless otherwise specified by the contracting officer, formation of lots and presentation of lots shall be in accordance with the "Submission of Product" provision of ASQ Z1.4.

**4.2.1.2 Size of lot.** A lot shall be of an identifiable group of assembled initiators containing metal parts assemblies produced by a homogeneous process and loaded with one lot of accepted cartridges.

**4.2.2 Examination.** One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4X) magnification may be used for inspection of such defects as burr and foreign matter. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristics shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected

**4.2.2.1 Cartridge, Impulse, M91, Drawing 8597230, covering a detail of Drawing 10520550.**

| Characteristic | Defect | Method of Inspection |
|---|---|---|
| **Critical** | | |
| 1 | Mixed cartridges | Visual |
| 2 | Metal defective | Visual |
| 3 | Marking missing, incorrect or illegible | Visual |
| **Major** | | |
| 101 | Foreign matter present | Visual |
| **Minor** | | |
| 201 | Case scratched, wrinkled, bulged, or dented | Visual |
| 202 | Varnish missing or inadequate | Visual |

**4.2.2.2 Assembly, Drawing 10520550, less all Drawings except 10520410, 10520411, and MS19060-3.**

| | | |
|---|---|---|
| **Critical** | | |
| 1 | Firing pin protrusion | Gage |



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| | | |
|---|---|---|
| 2 | Steel ball, missing | Gage |
| 3 | Striker hole in firing pin housing obstructed | Visual |
| **Major** | | |
| 101 | Burr | Visual |
| 102 | Foreign matter present | Visual |
| **Minor** | | |
| None defined | | |

**4.2.2.3 Assembly, Drawing 10520550, less Drawing 8597230, 10520407, 10520496, 10520414, and NAS 832-4.**

| | | |
|---|---|---|
| **Critical** | | |
| 1 | Firing pin not cocked | Manual/Visual |
| 2 | Any part missing | Visual |
| 3 | Improper assembly | Visual |
| **Major** | | |
| 101 | Safety pin loose or deformed | Visual |
| 102 | Burr | Visual |
| 103 | Foreign matter present | Visual |
| **Minor** | | |
| None defined | | |

**4.2.2.4 Assembly, Drawing 10520550, less all Drawings except 10520407, 10520414, 8597230.**

| | | |
|---|---|---|
| **Critical** | | |
| 1 | Cartridges not properly seated | Visual |
| 2 | Filter missing or insecure | Visual |
| 3 | Nylon patch missing or improperly applied | Visual |
| **Major** | | |
| 101 | Burr | Visual |
| 102 | Foreign matter present | Visual |
| **Minor** | | |
| None defined | | |

**4.2.2.5 Assembly Drawing 10520550.**

| | | |
|---|---|---|
| **Critical** | | |
| 1 | Cartridge face not seated against base of firing pin housing | X-ray |
| 2 | Any part missing except as otherwise classified | Visual/X-ray |
| 3 | Improper assembly | Visual/X-ray |
| 4 | Marking missing, incorrect, or illegible | Visual |
| **Major** | | |
| 101 | Setscrew, missing or insecure | Visual |
| 102 | Burr | Visual |
| 103 | Foreign matter present | Visual |
| **Minor** | | |
| 201 | Thread sealing compound, missing from setscrews | Visual |
| 202 | Protective cap missing | Visual |
| 203 | Protective finish, inadequate or defective | |



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**4.2.3 Inspection testing.** The following tests shall be performed in accordance with the provision of 4.3.

**4.2.3.1 Breakaway torque.** Each assembly shall be subjected to the breakaway torque test of the threaded joint between the cap and the chamber. Failure of any assembly to comply with the requirements of 3.3 shall be cause for rejection of the assembly.

**4.2.3.2 Residual magnetism.** Each initiator shall be subjected to the test for excessive residual magnetism as specified in 4.3.2. Failure on any initiator to comply with the requirement of 3.4 shall be cause for rejection.

**4.2.3.3 Firing pin release.** The initiator pin of each subassembly, consisting of all parts except the safety pin, cartridge and chamber, shall be subjected to the firing-pin release test. Failure of any subassembly to comply with the requirements of 3.5 shall be cause for rejection of the subassembly.

**4.2.3.4 Initiator pin security.** The initiator pin of each subassembly, consisting of all parts except the chamber and cartridge, shall be subjected to the initiator-pin security test. Failure of any subassembly to comply with the requirements of 3.6 shall be cause for rejection of the subassembly

**4.2.3.5 Functional tests.** The following tests shall be performed by the Government.

**4.2.3.5.1 Ballistics.** Twenty-one units shall be tested to determine compliance with the requirements of 3.7. The units shall be temperature conditioned as provided in table II and 4.3.5.2. Failure of any unit to comply with the requirements of 3.7 shall be cause for rejection of the lot.

**4.2.3.5.2 Retest.** There shall be no retest.

**4.2.3.5.3 Test failure.** If a test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and that test reconducted.

Table II. *Conditioning temperature.*

| Temperature (°F) | Sample Size |
|---|---|
| - 60 to - 70 | 7 |
| 65 to 75 | 7 |
| 195 to 205 | 7 |

**4.3 Test methods and procedures.**

**4.3.1 Breakaway torque.** A torque wrench set at the torque requirement specified in 3.3 shall be used in conjunction with a fixture to determine the adequacy of the joints. The chamber shall be secured in a suitable vise and the torque shall be applied to the cap.

**4.3.2 Residual magnetism.** The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass in a north-south horizontal position (cap end at south) with the compass set in the east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degrees rotation of the assembly for four full passes.



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**4.3.3 Firing pin release.** The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction, in conjunction with the fixture, which registers the amount of pull.

**4.3.4 Initiator pin security.** The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with a fixture, which registers the amount of pull. The test shall be performed with the safety pin in both positions.

**4.3.5 Functional.**

**4.3.5.1 Ballistics.** The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time is exceeded, the sample unit shall be reconditioned in accordance with 4.3.5.2. Required pressure readings shall be taken as specified in 3.7.1.

**4.3.5.2 Temperature conditioning.** Each assembled initiator shall be conditioned for a period of time not less than 4 hours nor greater than 24 hours. When reconditioning initiators at –60°F, all condensation shall be removed from the interior surfaces before returning the initiator to the conditioning chamber.

**4.4 Radiographic examination.** All assembled initiators shall be radiographically examined in accordance with ASTM E1742/E1742M, with exception that an initiator assembly will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The initiators shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the x-ray machine. All initiators shall be identified with serial numbers prior to x-ray examination. The serial numbers shall be in consecutive numerical order beginning with the number 001 in each production lot. Any missing serial numbers are to be identified on the x-rays. The initiators shall be arranged on boards or trays in consecutive order and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic plate identification shall include item nomenclature, the complete lot number, the span of serial numbers displayed, and the contract number. The radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective initiators found during x-ray review are to be marked on the x-ray prior to shipment, and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

| | | |
|---|---|---|
| a. | O-ring present | P/N 10520496 |
| b. | Steel balls present (3 ea) | P/N MS19060-3 |
| c. | Firing pin | P/N 10520411 |
| d. | Spring, initiator | P/N 10520412 |
| e. | Seal | P/N 10520415 |
| f. | Firing pin housing | P/N 10520410 |
| g. | Filter | P/N 10520414 |
| h. | Cap setscrews | (commercial) |
| i. | Cartridge, Impulse M91 | P/N 8597230 |
| j. | Initiator pin | P/N 10520553 |

**4.5 Inspection of packaging.** The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5.



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## 5. PACKAGING

**5.1 Packaging.** For acquisition purposes, the packaging requirements shall be as specified in the contractor order (see 6.2). When actual packaging of material needs to be performed by DOD personnel or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity

## 6. NOTES

(This section contains information of a general or explanatory nature that may helpful but is not mandatory.)

**6.1 Intended use.** The initiator assembly is used in conjunction with other propellant actuated devices in aircrew emergency escape systems. The initiator assembly was designed for use in a military aircraft and has no commercial application.

**6.2 Acquisition requirements.** Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. If required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Production test activity and lot size (see 4.1 and 4.2.1)
d. Direction for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.4).
e. Level of marking and packing (see 5.1 and 6.3).
f. Safety precautions (see 6.4).

**6.3 Contract packaging, packing and marking.** The following requirements should be specified in the contract.

**6.3.1 Packing and marking level.** Packing and marking should be Level A or B, as specified by the contracting activity.

**6.3.1.1 Level A.** Level A should be in accordance with Drawings 8830984 and 8830985 or SPI TPO F00-138-3770.

**6.3.1.2 Level B.** Level B should be in accordance with Drawings 8830984 and 9241155 or SPI TPO F00-138-3770.

**6.3.1.3 Lot numbers.** Lot numbers should be in accordance with MIL-STD-1168.

**6.4 Safety precautions.** The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by Federal Acquisition Regulation (FAR) 23.3. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the



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safety precaution requirements of “Ammunition and Explosives Ashore, ” OP 5, should be made applicable.

**6.5 Hazard notice.** The initiator described herein and certain of its components are flammable, explosive, or both. Consequently, they present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrical-induced incidents.

**6.6 Definitions.**

**6.6.1 Mechanical failure.** Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

**6.6.2 Ignition delay time.** Ignition delay time is the time between the firing pin striking the primer or ignition device and the first indication of pressure.

**6.7 Subject term (key word) listing.**

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

**6.8 Changes from previous issue.** Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes. These changes were authorized by Indian Head ECP 12E25BW007.

Custodian:
Navy-OS
Air Force-99

Preparing activity
Navy - OS
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Review activity:
Air Force-70

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